Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)
Degree, Semester & Branch: III Semester B.E. ECE A & B
Course Code & Title: EC8351 Electronic Circuits I
Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic: Unit 4 /** High frequency analysis of CE amplifier
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen: Collaborative Learning**
- **Justification:**
  Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve a problem, complete a task. Used to analyze the High frequency analysis of CE amplifier. Students must understand the concept of high frequency analysis and its importance.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**

First split the students into groups and ask them to sit as a group. The topic is divided into sub topics and each group of students allotted for sub topics. Students were asked to prepare the topic prior. They discussed within their group about the topic and present about the common Emitter amplifier and its high frequency analysis.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PO12	PSO3
CO4	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PO10	PSO3
	2	1	1
Justification for correlation	In assignment the technical information will be interpreted in basic level and the same will be presented as a report. So, it is mapped at level 2	Using advanced technology, students will show the frequency response of an amplifier. Thus the level 1 is mapped	Students will design an amplifier to achieve the desired frequency response. Hence level 1 is mapped.

- **Images / Screenshot of the practice:**

 

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students stated that they appreciated the activity and that it helped them remember topics and content when writing assessments.

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    The Students eagerly participated in that activity and then they acquired the concept of High frequency analysis of CE amplifier. Due to this activity, the students were able to recall the topic and remember the important terms easily.

  - ***Challenges faced in implementation:***
    I have planned this activity for 40 minutes. Students take some time to forming the group. After formed the group, they discussed the topic. Students asked doubts regarding the topic before presentation. I've given a brief overview of the amplifier. I encouraged the students which it takes another 5 minutes for completing the activity.

References:

- Salivahanan and N. Suresh Kumar, Electronic Devices and Circuits, 4th Edition, , McGrawHill Education (India) Private Ltd., 2017.

- Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010

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Degree, Semester & Branch: III Semester B.E. ECE A & B
Course Code & Title: EC8351 Electronic Circuits I
Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic: Unit 5 /** Regulated Power Supply
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 17
- **Activity Chosen: Flipped Classroom**
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of "Regulated power supply". They described the operation of a "regulated power supply". Students were divided into groups and each group will prepared the detailed presentation of Regulated power supply and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO3
CO5	2	2	3	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PO12	PSO3
	2	1	2
Justification for correlation	Students will understand the concept, design of Rectifier & RPS circuit and will give clear information during discussion session. Hence it is mapped at level 2	Students were identify the previous instances in Rectifier designing where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus	Students will design Rectifier circuit with and without filter. Hence it is mapped at level 2

		level 1 is mapped	

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

    The Students eagerly participated in that activity and then they acquired the knowledge of regulated DC Power supply. Due to this activity, the students were able to recall the topic and remember the important terms easily.

  - ***Challenges faced in implementation:***

    I have planned this activity for 30 minutes. Students take some time to present the topic. After formed the group, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the regulated power supply and encouraged the students to present which it takes another 5 minutes for completing the activity.

References:

- Salivahanan and N. Suresh Kumar, Electronic Devices and Circuits, 4th Edition, , McGrawHill Education (India) Private Ltd., 2017.

- Donald. A. Neamen, Electronic Circuits Analysis and Design, 3rd Edition, Mc Graw Hill Education (India) Private Ltd., 2010

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Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic : Thermal stability, Stability factors
Innovative Method : One Minute Paper

One Minute Paper

31/8/21

1) Other Name for Q-point

Quiescent point, operating point, bias point

2) Q-point value

$Q = (I_C, V_{CE})$

3) Stability factors

S, S', S''

4) Fixed bias also called as

Base Bias

5) Stability factor 'S' for Voltage divider bias current

2. $S = \frac{\Delta I_C}{\Delta I_{CO}}$

$\boxed{S = 2}$

K. Manoj
II - ECE (A)
953621106041

1) Other names for Q point

Quiescent point, Bias point, operating point

2) Q point value

$Q = (I_C, V_{CE})$

3) Stability factors

S, S', S''

4) Fixed bias is also called

Base bias

5) Stability factor "S" for voltage divider circuit

2.

Reg No: 953621106063
Name: R.Ranjith
Class: ECE-B

1) Other name for Q-point is Bias Point or/ Quiescent Point

2) Q-point value:

0.01%

3) Stability Factor:

The degree of change in operating point due to variation in temperature

4) Base Bias

5) $S = \frac{dI_C}{dI_{CO}} \Big|_{V_{BE}, \beta\ (constant)}$

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1. Other name of Q-point is bias point.

2. Q = 6.6 mA

$(V_{CE}, 0) - (0, \frac{V_{CE}}{R_C})$

3. It is the degree of change in operating point due to variation in temperature.

4. Fixed bias is also called as Base bias

5. Stability factor (S) is defined as the rate of change of collector current (I_C) with respect to reverse saturation current (I_{CO}) keeping β and V_{BE} constant.

$S = \frac{dI_C}{dI_{CO}}$ at constant β and V_{BE}

Minimum value of 'S' is '1' for small values of (R_{TH}/R_E)

Maximum value of 'S' is 1+β for large (R_{TH}/R_E)

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Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Various biasing methods of JFET - JFET Bias Circuit Design
Innovative Method **:** Brain Storming

	
	

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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Various biasing methods of JFET - JFET Bias Circuit Design
Innovative Method **:** Class Poll

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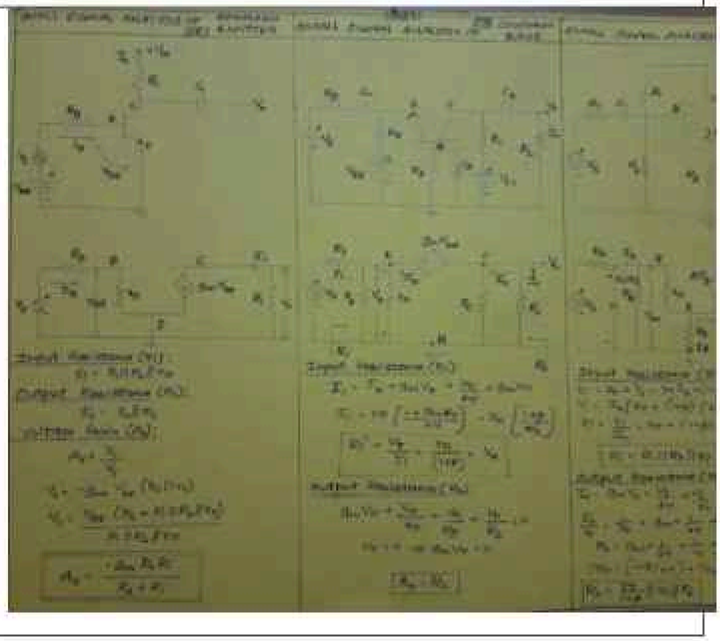
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic : Analysis of common base (CB) amplifier using Hybrid π equivalent circuits
Innovative Method : Chart Preparation



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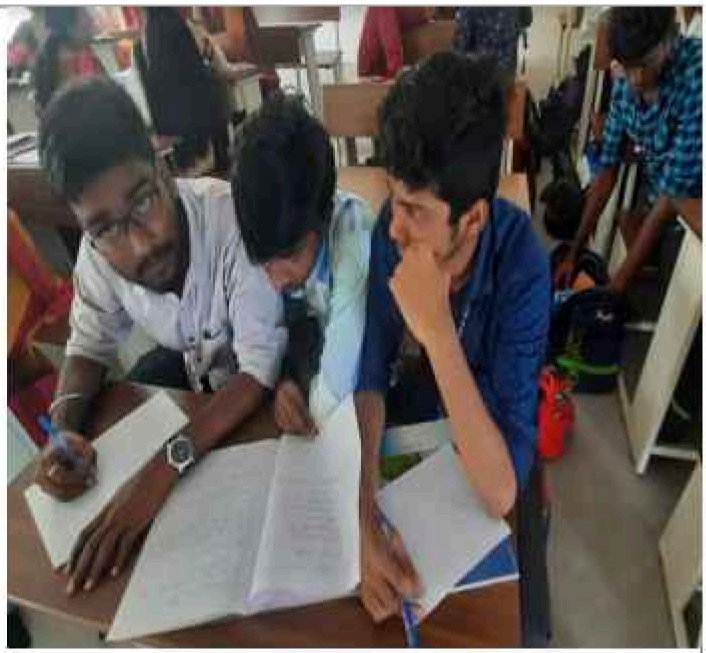
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Differential amplifier, Basic BJT differential pair
Innovative Method **:** Mind Map



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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic : Small signal analysis and CMRR
Innovative Method : Open Book Test

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Department of Electronics and Communication Engineering
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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Analysis of JFET Common Gate amplifier using Hybrid π equivalent circuits
Innovative Method **:** Mind Map




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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Analysis of MOSFET Common Gate amplifier using Hybrid π equivalent circuits
Innovative Method **:** Chart Preparation




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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** BiCMOS circuits – Basic amplifier stage, current source
Innovative Method **:** Think Pair Share




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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Amplifier frequency response
Innovative Method **:** Brain Storming

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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A
Topic : Transistor Switching Times
Innovative Method : One minute paper

953620106013 1. Delay time 2. Rise time 3. $t_r = T_{ON} + T_S$ 4. 10% to 90%	953620106002 1) Delay Time 2) Rise Time 3) $T_r = T_{on} + T_s$ 4) 10% to 90%
953620106080 1. Delay time 2. Rise time 3. $T_r = T_{ON} + t_s$ 4. 10% to 90%	953620106035 1 Delay time 2 Rise time 3. $t_r = T_{ON} + t_s$ 4. 10% to 90%

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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Full-Wave Rectifier Power Supply
Innovative Method **:** Demonstration in Laboratory

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Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC8351 Electronic Circuits I
Semester, Degree & Branch : III B.E ECE A & B
Topic **:** Switched mode power supply (SMPS)
Innovative Method **:** Open Book analysis